RITIK POKHAREL

GIS ANALYST | ENTERPRISE GIS | UAV REMOTE SENSING | GEOAI

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PROFESSIONAL SUMMARY

GIS Analyst with an M.S. specializing in GIS and remote sensing who delivers production Esri solutions across enterprise administration, offline field operations, web applications, spatial analysis, and technical training. Combines ArcGIS Pro, Enterprise, Python, UAV photogrammetry, and machine learning to turn complex spatial data into reliable client and field workflows. First author in *Remote Sensing* with experience communicating results to stakeholders, students, and GIS professionals.

TECHNICAL SKILLS

GIS & Esri: ArcGIS Pro, ArcGIS Enterprise (Portal, Server, hosted feature services), ArcGIS Online, Experience Builder, Dashboards, StoryMaps, Survey123, Field Maps, ModelBuilder, Spatial Analyst, Image Analyst, QGIS

Programming & Data: Python (ArcPy, ArcGIS API for Python, pandas, NumPy, GeoPandas, scikit-learn), R, SQL/PostgreSQL, Git/GitHub, Jupyter, Arcade, JavaScript, file and enterprise geodatabases, data migration, QA/QC, metadata

Remote Sensing & GeoAI: UAV multispectral acquisition, Pix4D Mapper, ArcGIS Drone2Map, orthomosaics, DSM/DTM, radiometric calibration, GCPs, NDVI/NDRE, Sentinel-2, Landsat, NAIP, supervised classification, change detection

Machine Learning: Random Forest, XGBoost, gradient boosting, neural networks (TensorFlow/Keras), SVM, PLSR, Ridge/LASSO/Elastic Net, feature engineering, cross-validation, independent-season validation

Field & Cartography: GNSS/GPS data collection, offline mobile deployment, map series, coordinate systems and projections, topology, symbology, print-ready layouts

PROFESSIONAL EXPERIENCE

GIS Analyst — Global Geospatial Institute / Global Geospatial Solutions, LLC

Jun 2026–Present

Baton Rouge, LA | Geospatial consulting, education, and workforce development

- Administer **ArcGIS Enterprise and a geospatial data server for two client organizations**, publishing hosted feature layers, managing access and sharing, and providing managed hosting for organizations without their own Enterprise environment.
- Built a parish-scale, offline **ArcGIS Field Maps disaster-response system** for Terrebonne Parish Consolidated Waterworks: 16 downloadable areas, approximately 10 operational layers, and 9–12 6K quad grid cells per area (about 4 sq. mi. each) for disconnected field access.
- Developed the **Gardere Community Hub** in Experience Builder and Survey123, integrating four operational layers and 60+ community resources; presented the deployed application to project stakeholders in Oct 2026. App: l.ead.me/gardereapp
- Administer an **ArcGIS Online organization with 2,000+ users and 500+ groups**; automated bulk account provisioning and role reassignment with the ArcGIS API for Python, replacing repetitive per-account steps with a scripted batch workflow.
- Designed and delivered hands-on ArcGIS Pro, ArcGIS Online, Survey123, and Field Maps training to **300+ students, 15 teachers, and four GIS professionals**; built reusable exercise datasets and step-by-step lab guides for K–12 and workforce programs.
- Delivered a GIS Maturity Assessment across **50 weighted criteria**, translating findings into an executive summary and prioritized implementation roadmap; also produced client-ready maps, map books, dashboards, StoryMaps, newsletters, and stakeholder presentations.

Graduate Research Assistant — Louisiana State University / LSU AgCenter

May 2024–May 2026

Baton Rouge, LA

- Planned UAV multispectral campaigns across three growing seasons (2023–2025), capturing **720 rice research plots at approximately 10-day intervals** under six nitrogen rates and multiple cultivars.
- Processed imagery in Pix4D Mapper into radiometrically calibrated, georeferenced **2.5 cm ground-sample-distance orthomosaics** using ground control points and reflectance-panel calibration; derived 19 vegetation indices for yield modeling.
- Benchmarked **12 machine-learning models** in Python, achieving $R^2 = 0.89$ in validation; XGBoost maintained $R^2 = 0.63$ and normalized RMSE below 9% on an independent 2025 season.
- Built reproducible Python and R workflows for multi-season field, sensor, and geospatial data; translated methods and findings into a first-author journal article, conference presentations, maps, and field-day communication.

Graduate Teaching Assistant — Louisiana State University — Jan 2024–May 2024

Baton Rouge, LA

- Instructed 50+ undergraduates across weekly BIOL 1208 laboratories, delivering protocols, assessment, and one-on-one technical support.

Research Manager — Gorkha Agri Group — Dec 2021–Dec 2023

Nepal

- Directed disease diagnostics and seed-quality control for commercial rice and hybrid tomato production; coordinated with NSAF, CIMMYT, and SEAN on certified seed release.

SELECTED PROJECTS

- **Rice Yield Estimation Using UAV and Machine Learning —** Reproducible Python workflow for UAV vegetation-index processing, model training across 300 iterations, independent-season evaluation, feature importance, and result visualization. View code
- **Gardere Community Hub —** Public-facing Experience Builder and Survey123 application connecting residents to 60+ mapped community resources across four operational layers. App: l.ead.me/gardereapp

EDUCATION

M.S., Plant, Environmental and Soil Sciences — Louisiana State University — May 2026

Baton Rouge, LA | GIS & Remote Sensing specialization; Statistics minor

Thesis: *Digital Agriculture for Rice Nutrient Management and Yield Prediction Using UAV-Based Remote Sensing and Machine Learning* | repository.lsu.edu/gradschool_theses/6375

B.S., Agriculture — Tribhuvan University, IAAS Paklihawa — 2021

Nepal

SELECTED PUBLICATIONS

- Pokharel, R., Gentimis, T., Kongchum, M., Tubana, B., Adhikari, R., & Setiyono, T. (2026). “Rice Yield Estimation Based on Machine Learning Applied to UAV Remote Sensing Data.” Remote Sensing, 18(10), 1575. doi:10.3390/rs18101575
- Baker, S., Islam, M. N., Bhandari, M., et al., including Pokharel, R. (2026). “Strategically Timed UAS Data Acquisition for Wheat Yield Prediction Using Explainable AI.” Smart Agricultural Technology, 15, 102606. doi:10.1016/j.atech.2026.102606
- Pokharel, R., Adhikari, R., Nepal, N., & Budhathoki, R. (2023). “Soil Analysis GIS-Based Fertility Assessment and Soil Mapping of Cardamom Growing Area, Pandam, Ilam, Nepal.” IJRASET. Article

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

In progress: GISP-E (PreGISP) examination — scheduled Nov 22, 2026; FAA Part 107 Remote Pilot preparation — certification not yet held

Esri MOOCs (4): Make an Impact with Enterprise GIS; The ArcGIS Imagery MOOC: Foundations and Frontiers; Spatial Data Science: The New Frontier in Analytics; The Power of Where: Communicating with Maps

Esri training certificates (11 total; selected): Experience Builder Basics; Introduction to ArcGIS API for Python; Creating and Sharing GIS Content Using ArcGIS Online; ArcGIS Online Basics; Telling Stories with GIS Maps; Getting Started with Imagery and Remote Sensing; ArcGIS Drone2Map Basics

Additional training: DSSAT Crop Modeling and Decision-Support Systems — University of Georgia, 2025

AWARDS & PRESENTATIONS

Third Place, Research Poster — AI in Agriculture and Natural Resources Conference, Mississippi State University, 2025 | **Oral Presentation** — ASA-CSSA-SSSA International Annual Meeting, Salt Lake City, 2025